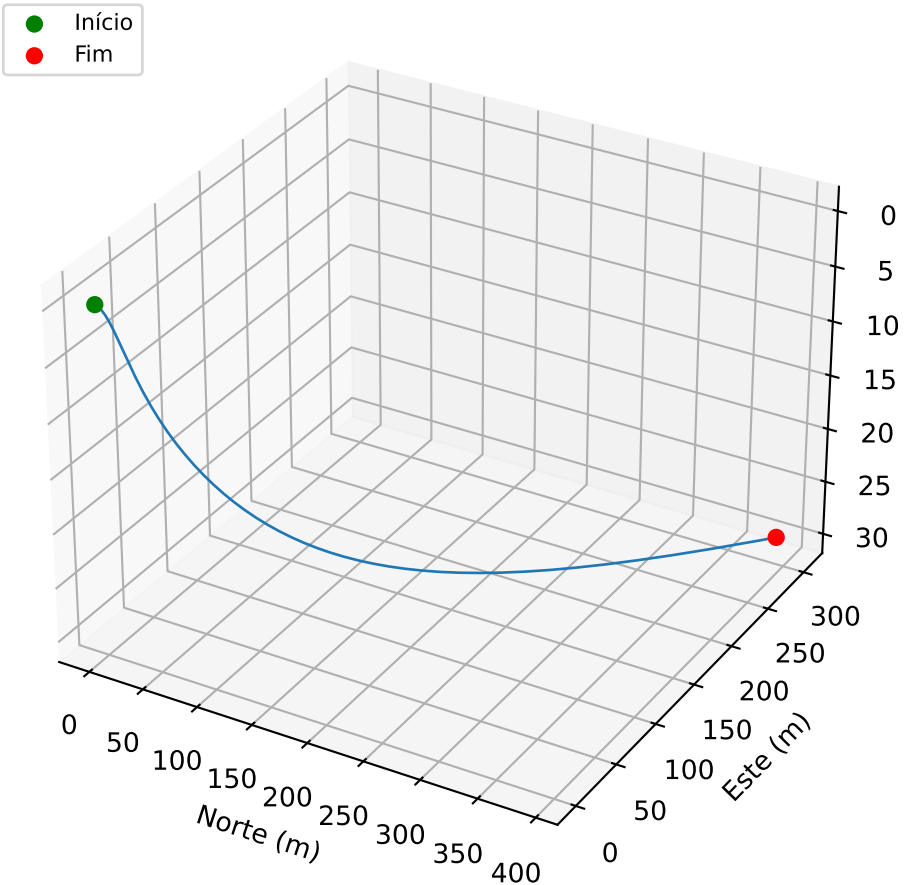
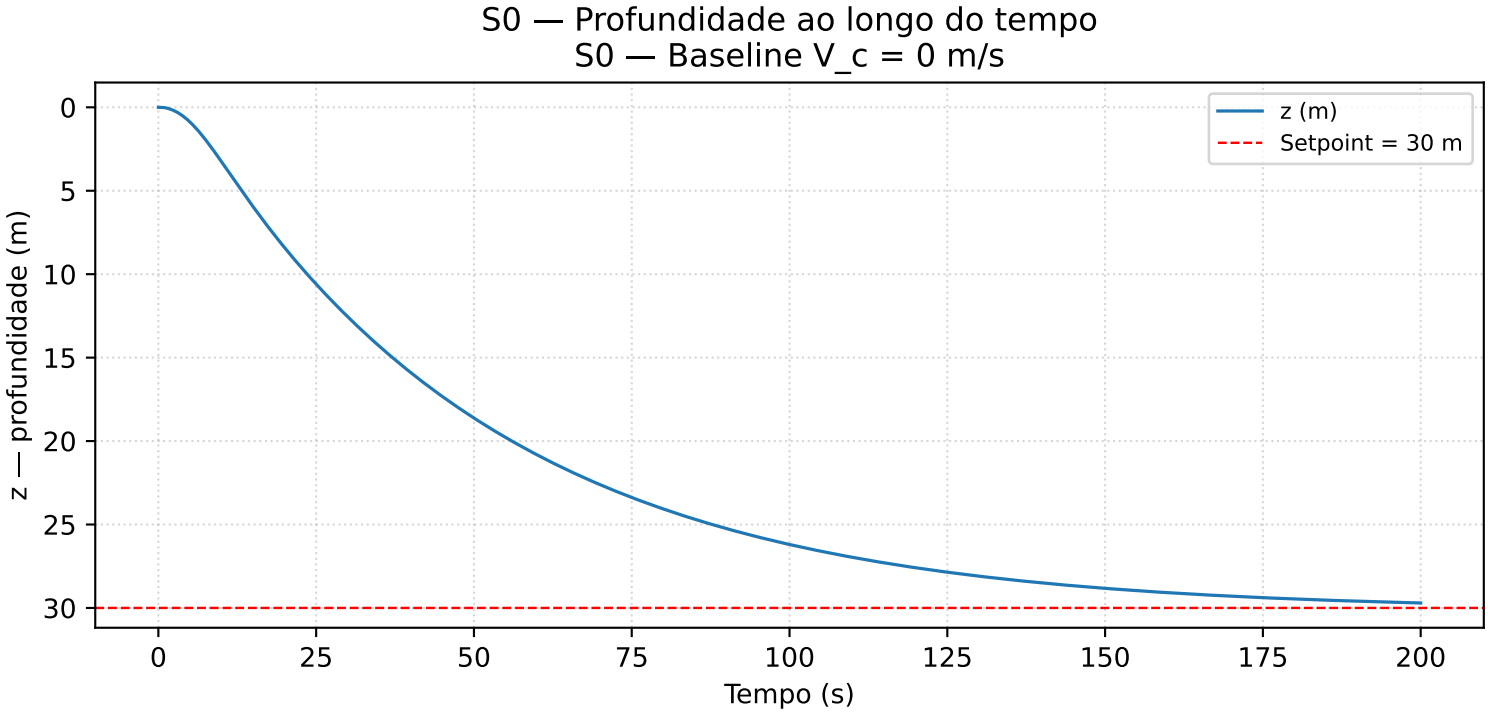
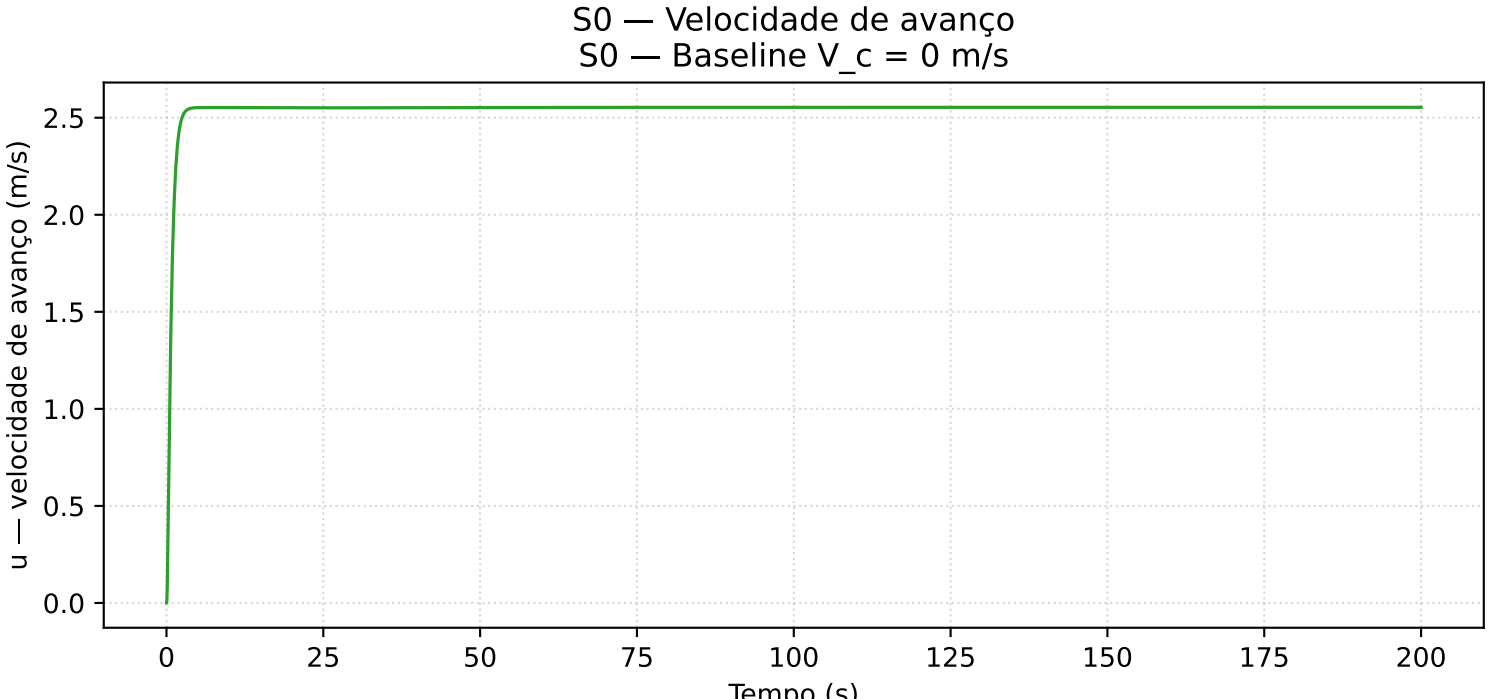
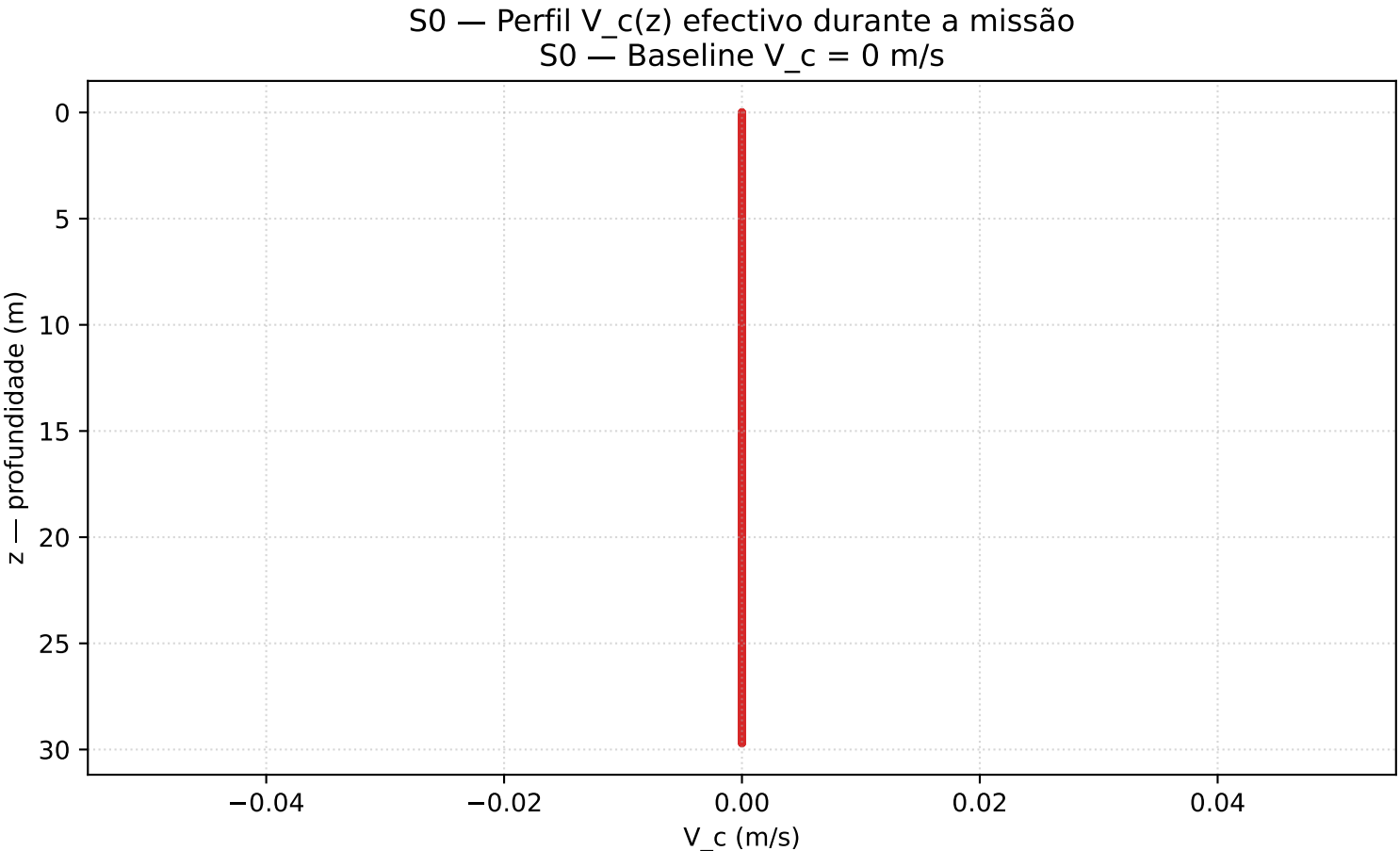
S0 — Trajectória 3D
S0 — Baseline V_c = 0 m/s

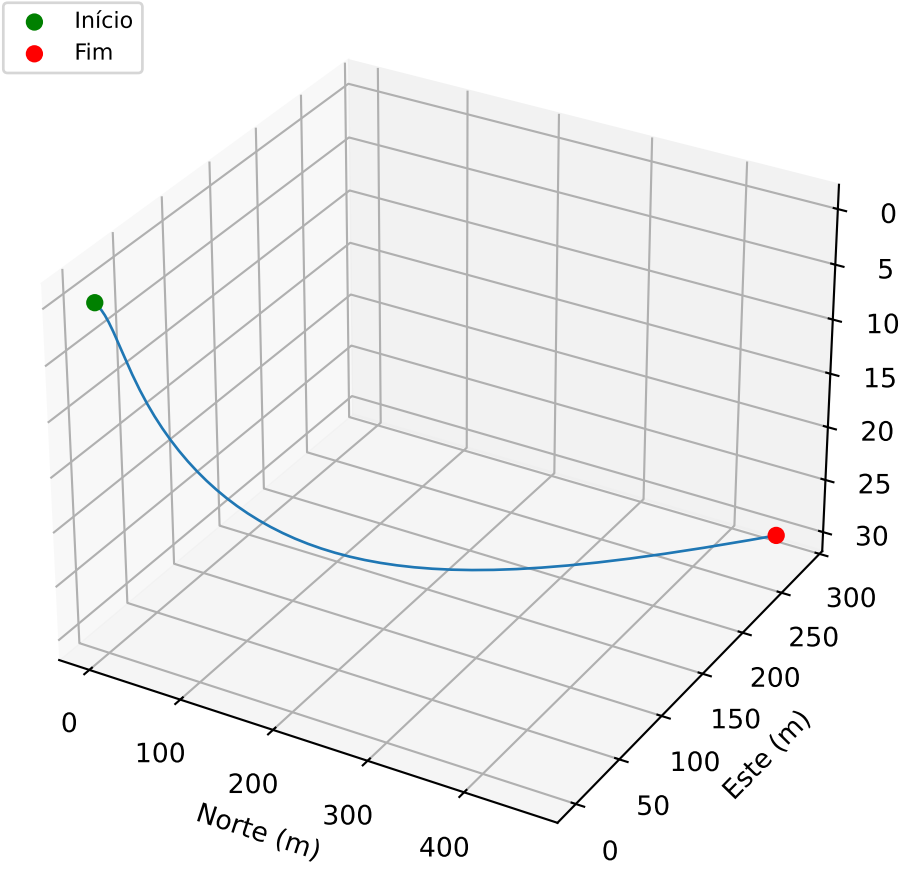


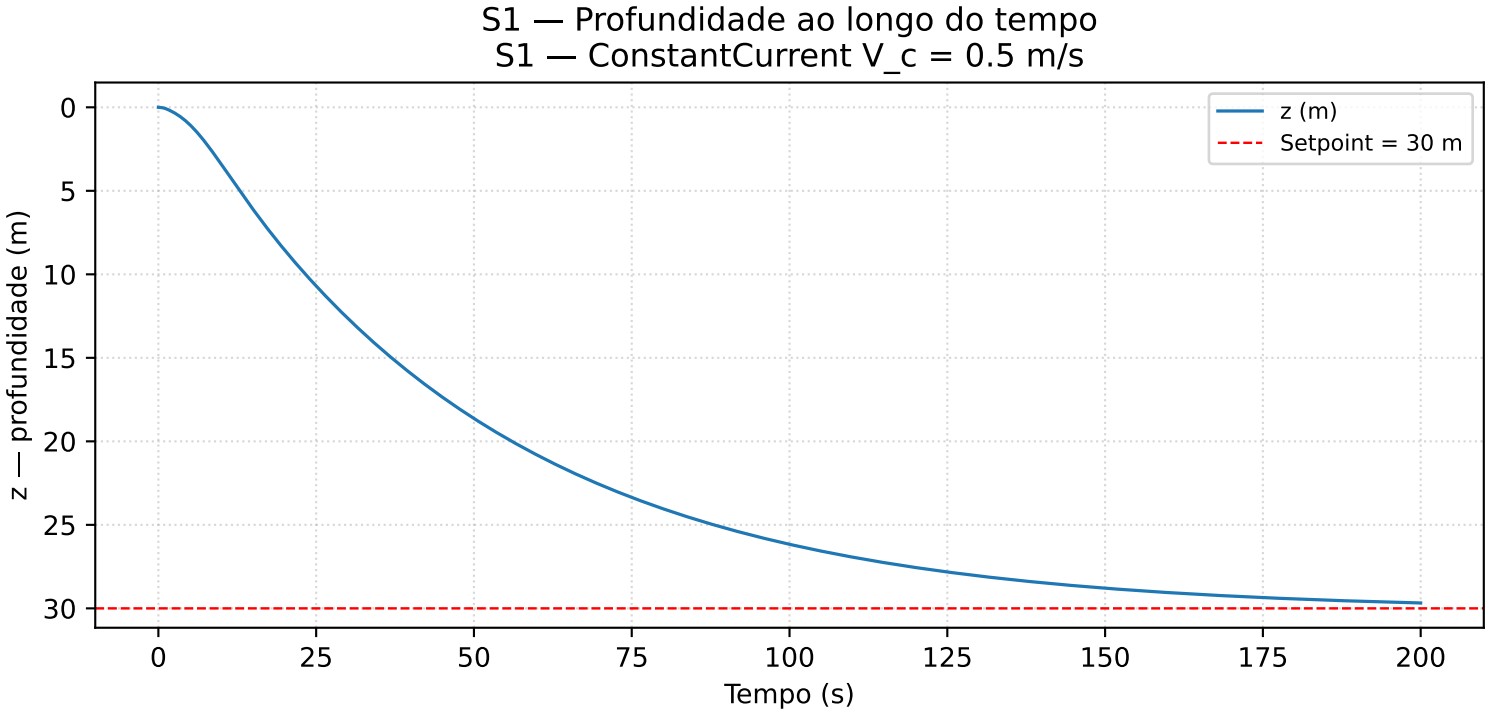

S0 — Profundidade ao longo do tempo
S0 — Baseline V_c = 0 m/s
z (m)
Setpoint = 30 m
z — profundidade (m)
Tempo (s)
0
5
10
15
20
25
30
0
25
50
75
100
125
150
175
200

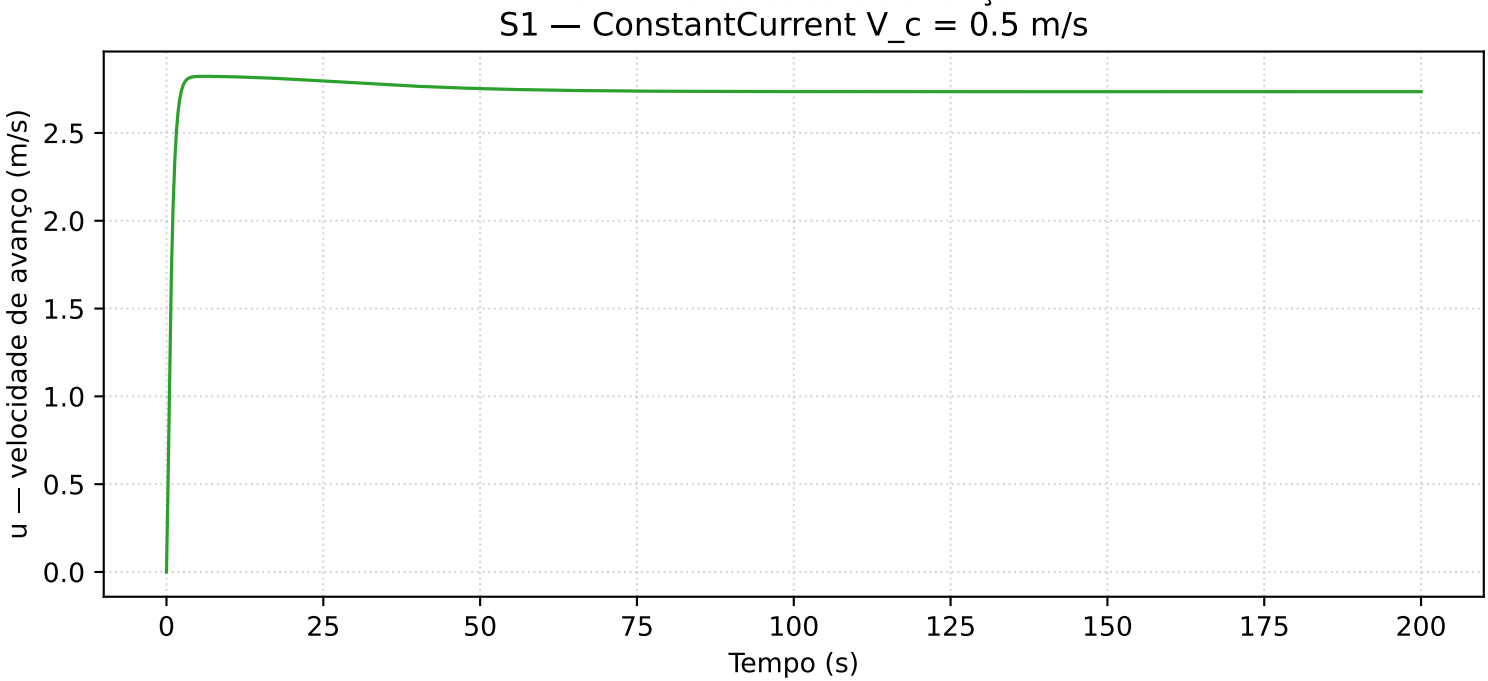

S0 — Velocidade de avanço
S0 — Baseline V_c = 0 m/s
u — velocidade de avanço (m/s)
Tempo (s)
0.0
0.5
1.0
1.5
2.0
2.5
0
25
50
75
100
125
150
175
200

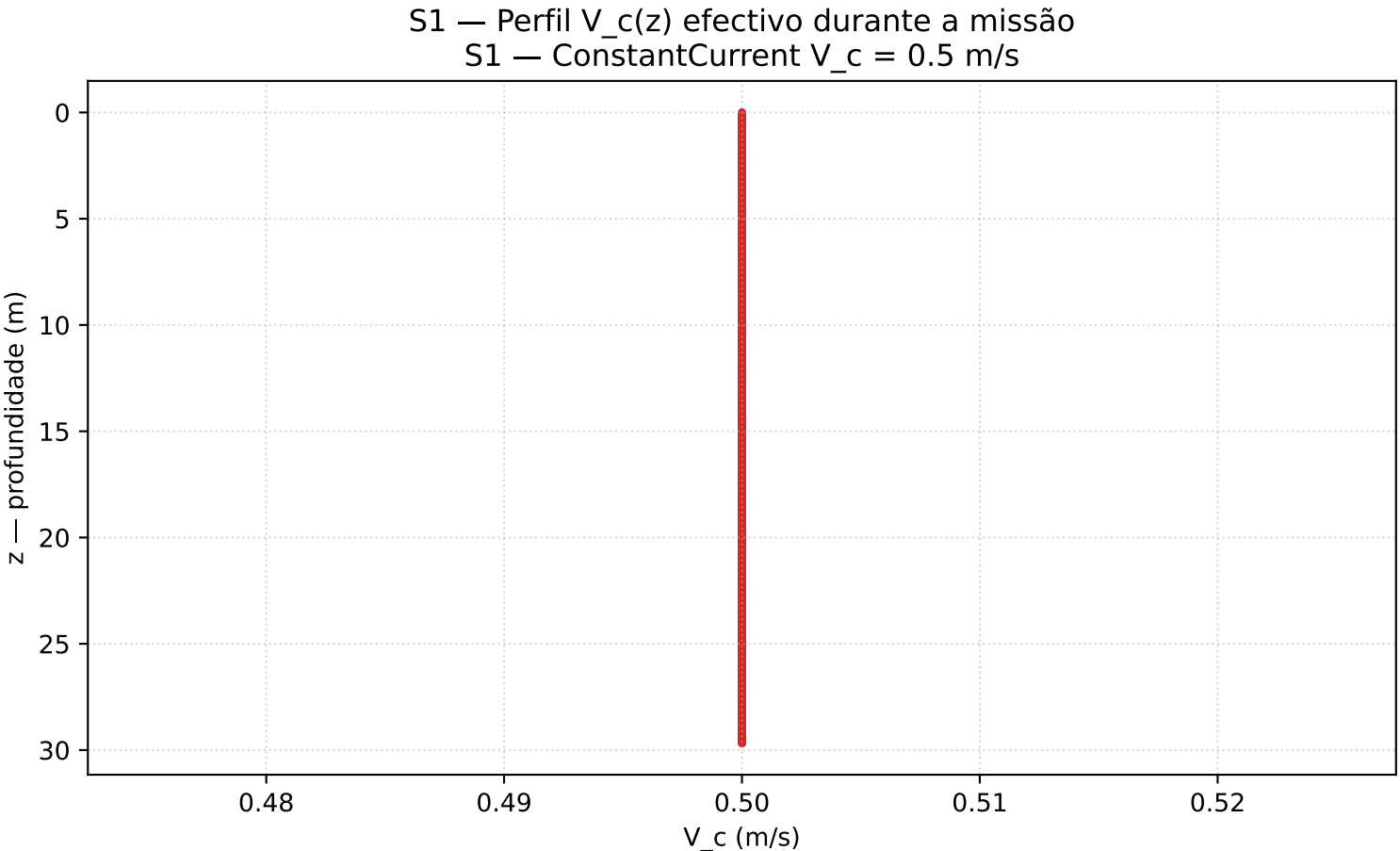

S0 — Perfil V_c(z) efectivo durante a missão
S0 — Baseline V_c = 0 m/s
z — profundidade (m)
0
5
10
15
20
25
30
−0.04
−0.02
0.00
0.02
0.04
V_c (m/s)

S1 — Trajectória 3D
S1 — ConstantCurrent V_c = 0.5 m/s

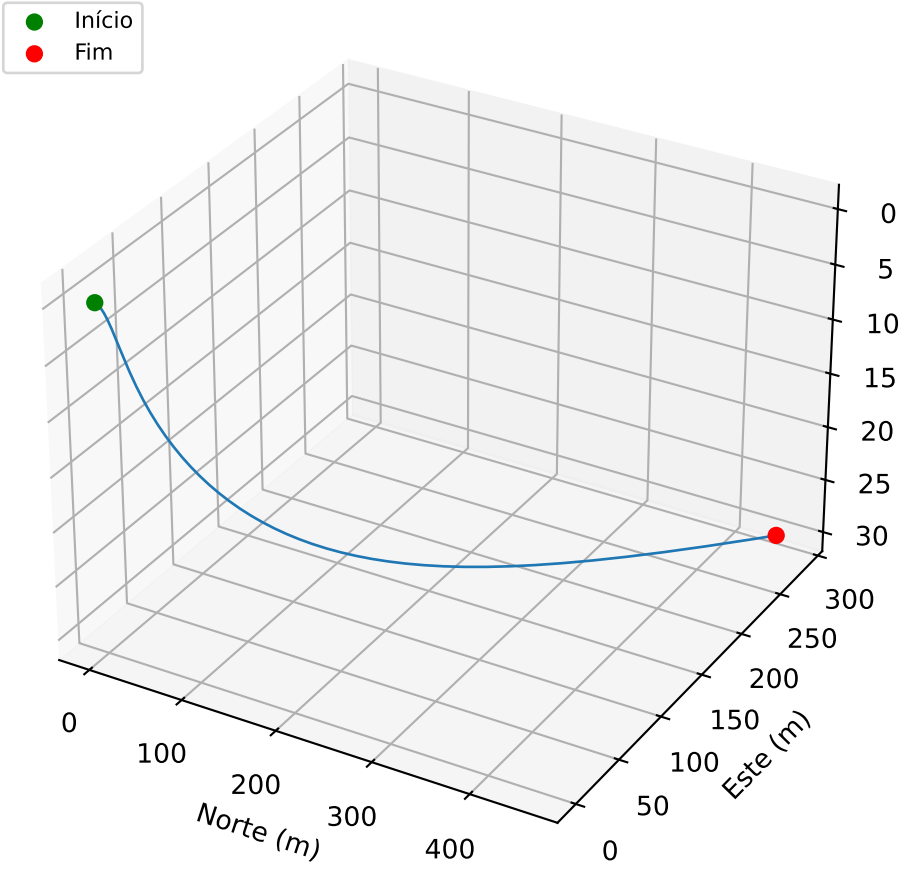


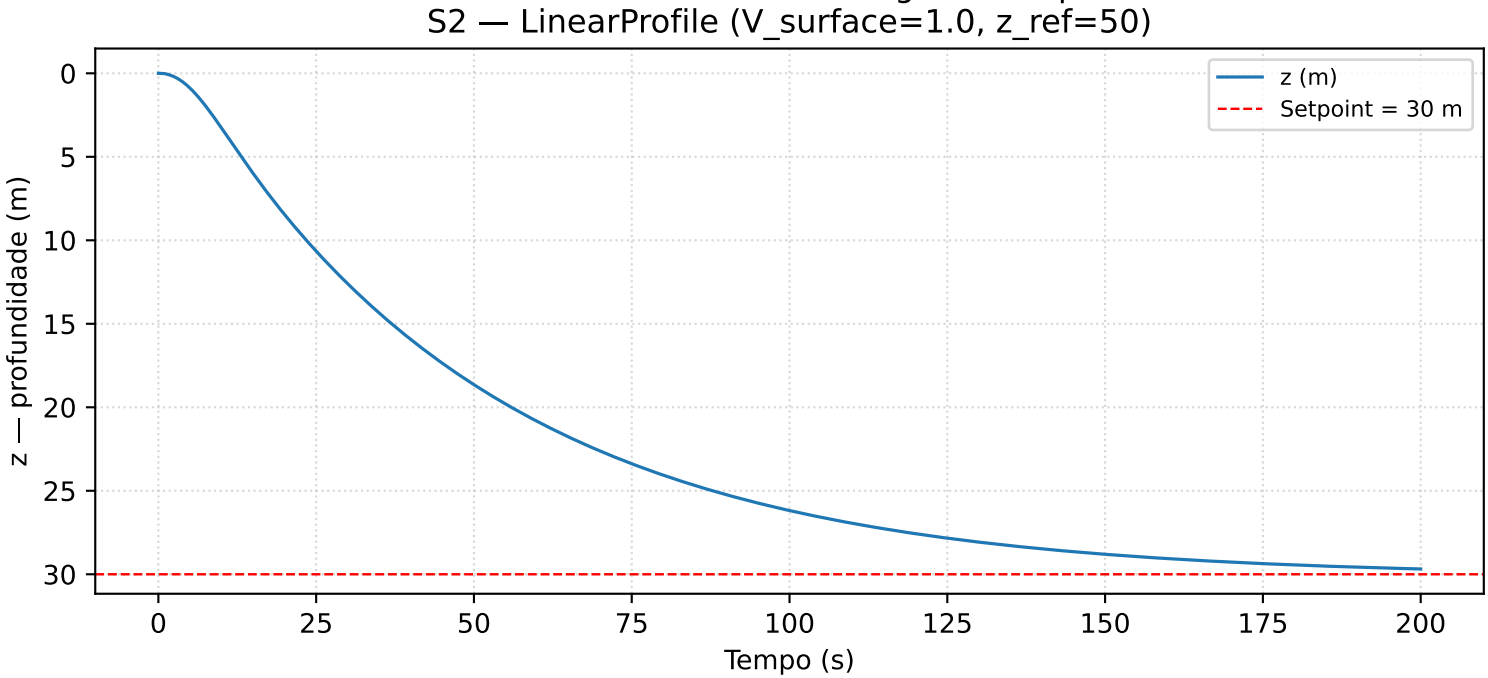

S1 — Profundidade ao longo do tempo
S1 — ConstantCurrent V_c = 0.5 m/s
z (m)
Setpoint = 30 m
z — profundidade (m)
Tempo (s)
0
5
10
15
20
25
30
0
25
50
75
100
125
150
175
200

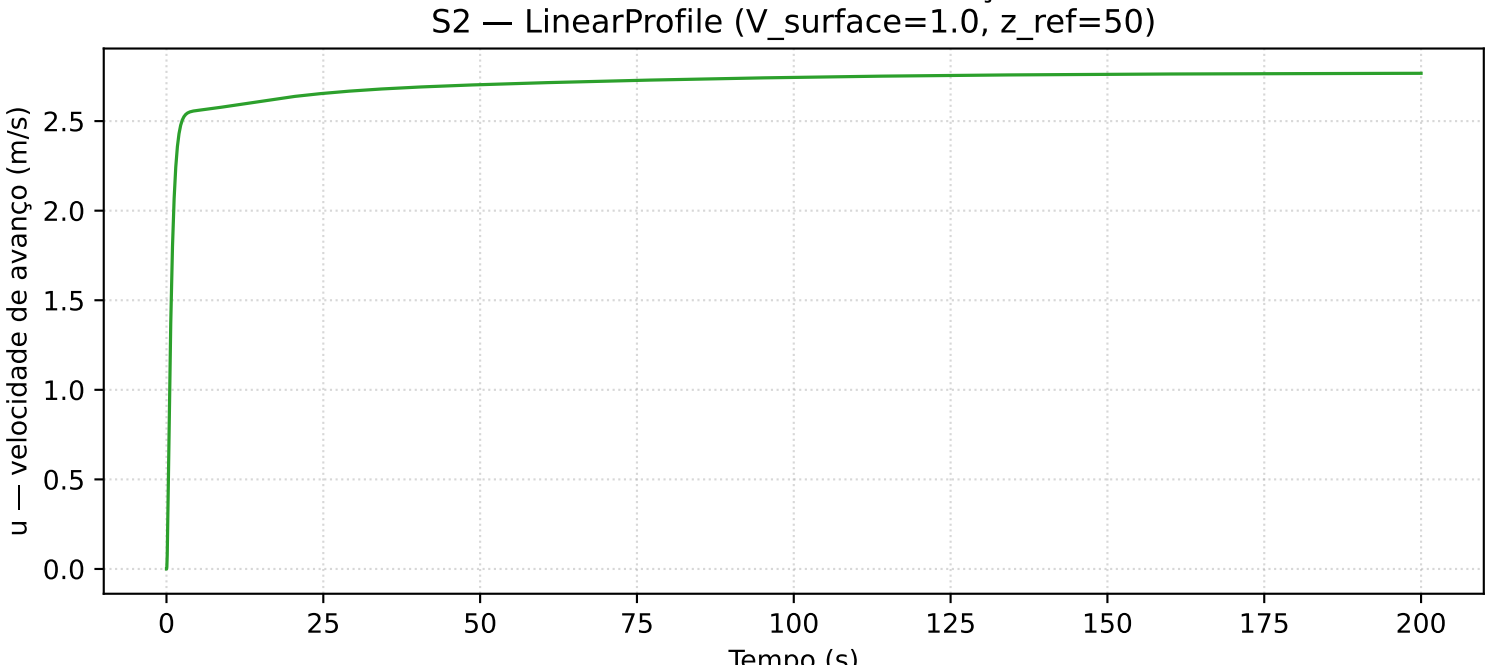

S1 — Velocidade de avanço
S1 — ConstantCurrent V_c = 0.5 m/s
u — velocidade de avanço (m/s)
2.5
2.0
1.5
1.0
0.5
0.0
0
25
50
75
100
125
150
175
200
Tempo (s)

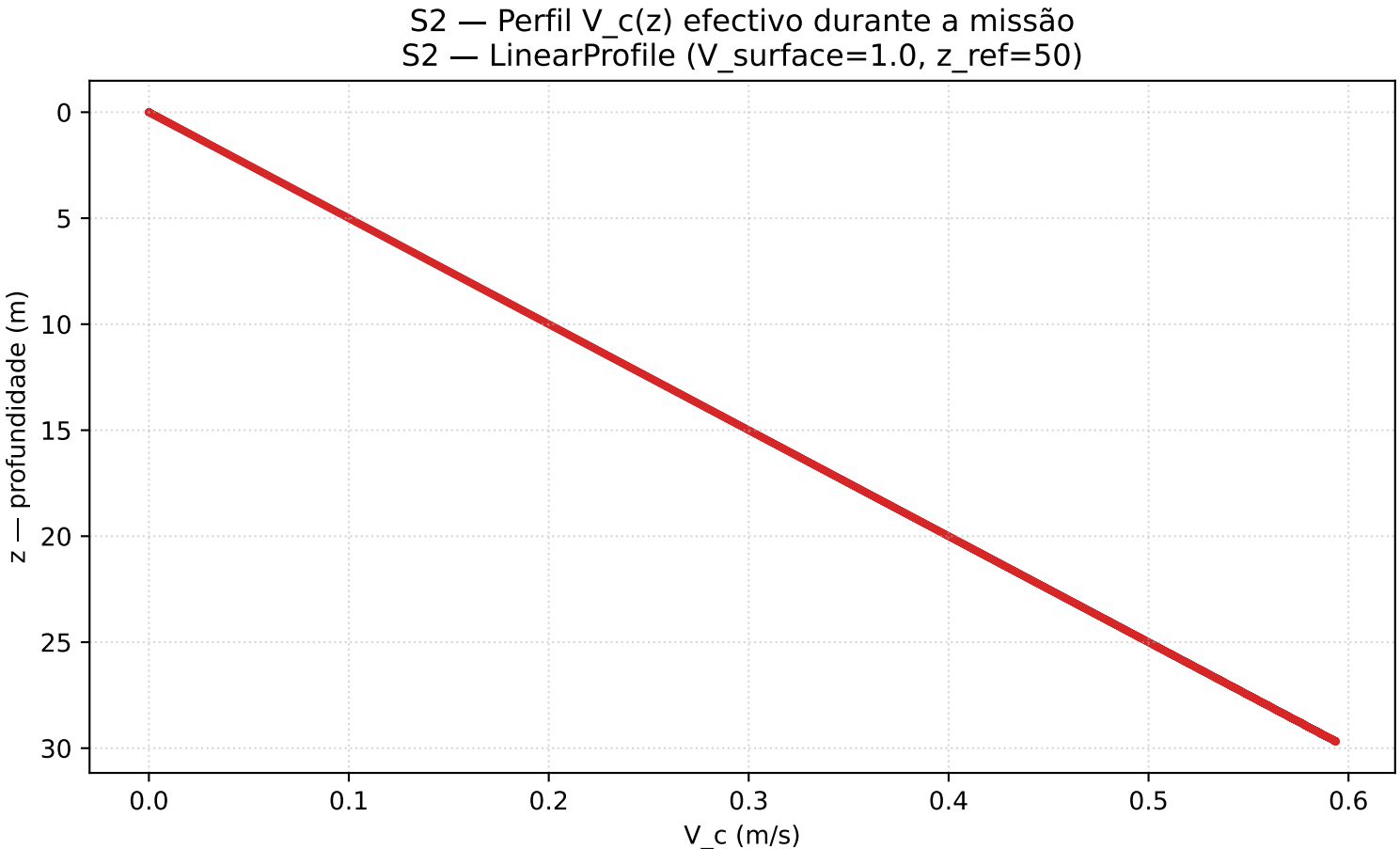

S1 — Perfil V_c(z) efectivo durante a missão
S1 — ConstantCurrent V_c = 0.5 m/s
z — profundidade (m)
0
5
10
15
20
25
30
0.48
0.49
0.50
0.51
0.52
V_c (m/s)

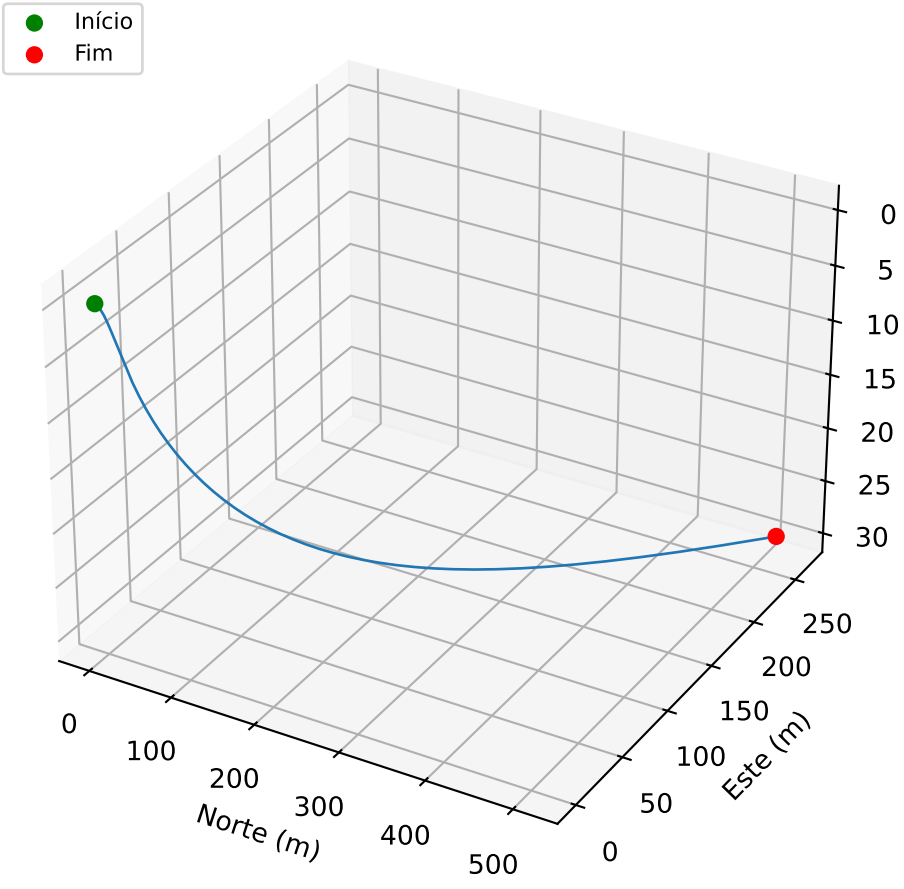
S2 — Trajectória 3D
S2 — LinearProfile (V_surface=1.0, z_ref=50)

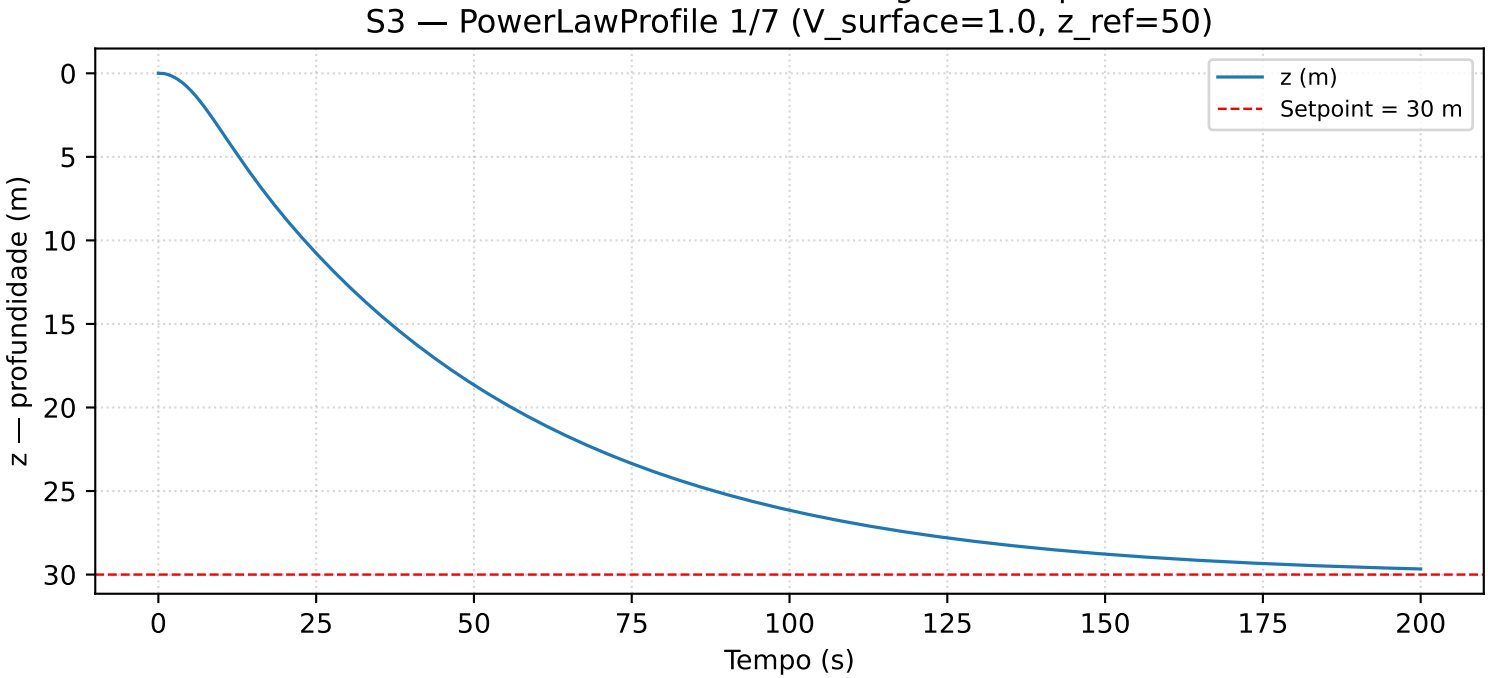


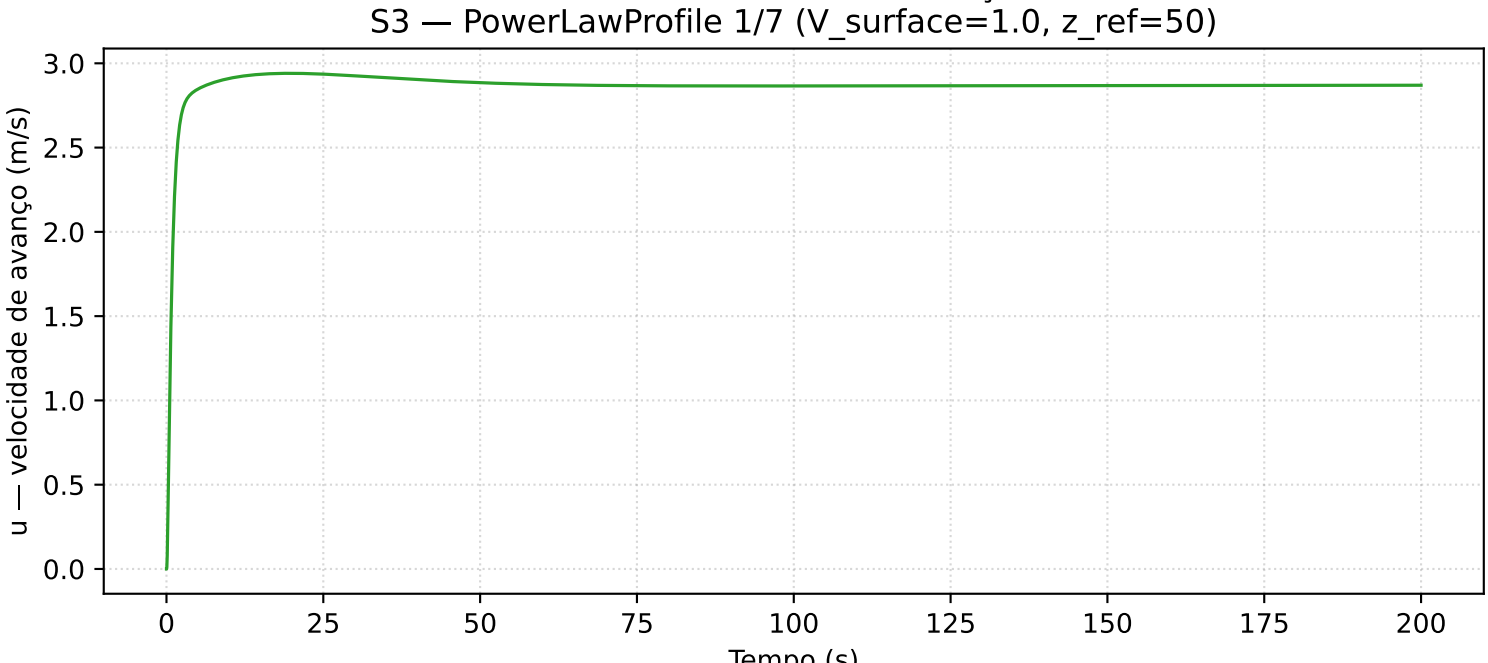

S2 — Profundidade ao longo do tempo
S2 — LinearProfile (V_surface=1.0, z_ref=50)
z (m)
Setpoint = 30 m
z — profundidade (m)
Tempo (s)
0
5
10
15
20
25
30
0
25
50
75
100
125
150
175
200

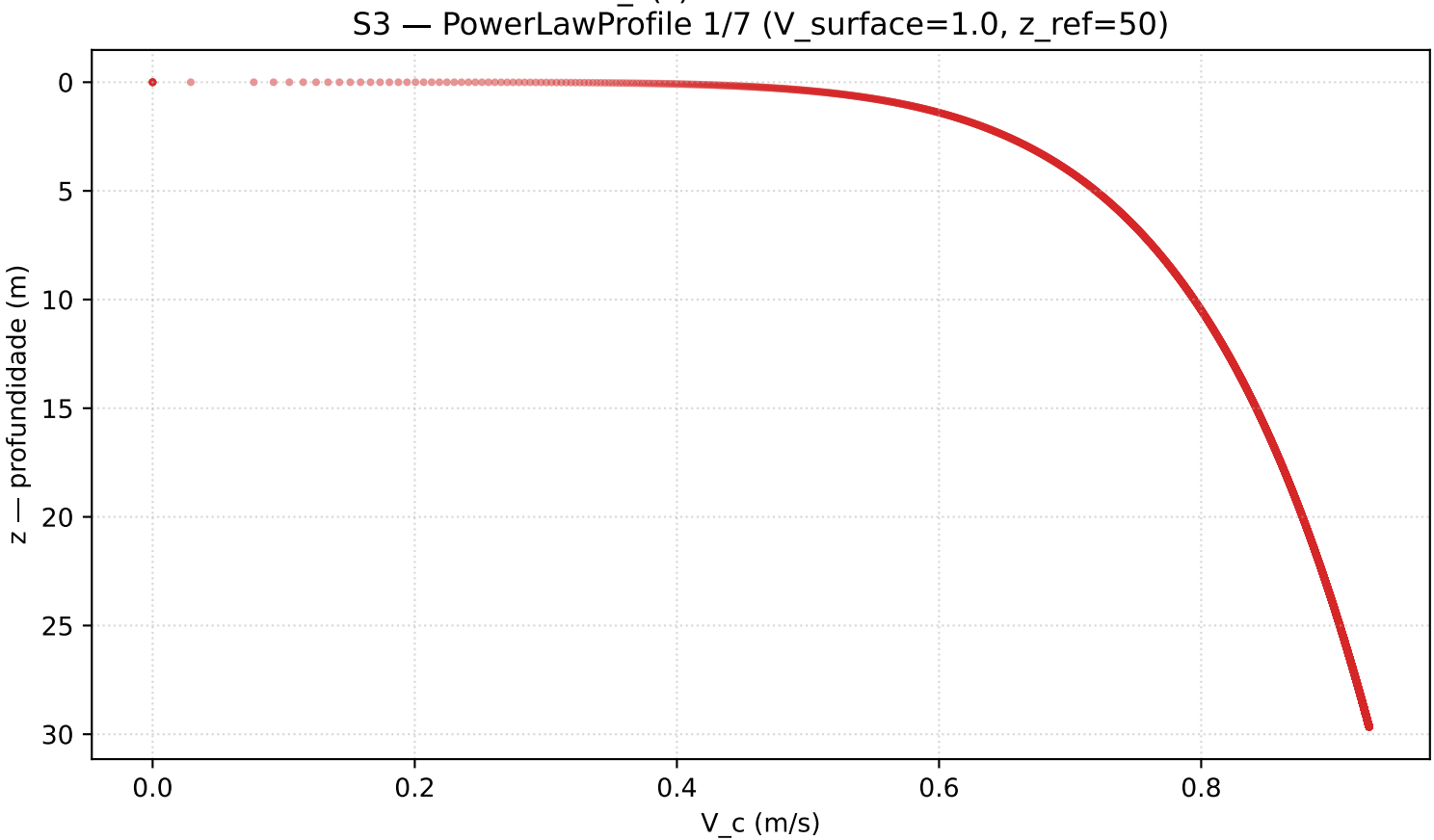

S2 — Velocidade de avanço
S2 — LinearProfile (V_surface=1.0, z_ref=50)
u — velocidade de avanço (m/s)
Tempo (s)
0.0
0.5
1.0
1.5
2.0
2.5
0
25
50
75
100
125
150
175
200


S2 — Perfil V_c(z) efectivo durante a missão
S2 — LinearProfile (V_surface=1.0, z_ref=50)
z — profundidade (m)
0
5
10
15
20
25
30
0.0
0.1
0.2
0.3
0.4
0.5
0.6
V_c (m/s)

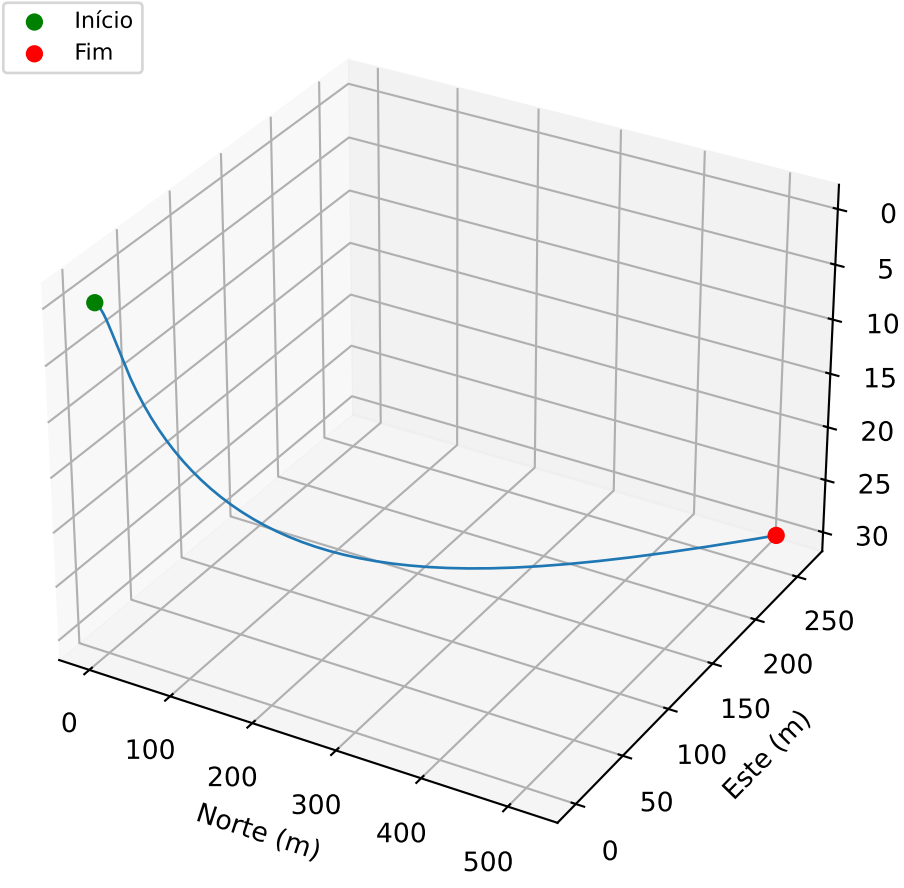

S3 — Trajectória 3D
S3 — PowerLawProfile 1/7 (V_surface=1.0, z_ref=50)
Início
Fim
0
100
200
300
400
500
Norte (m)
0
50
100
150
200
250
Este (m)
0
5
10
15
20
25
30

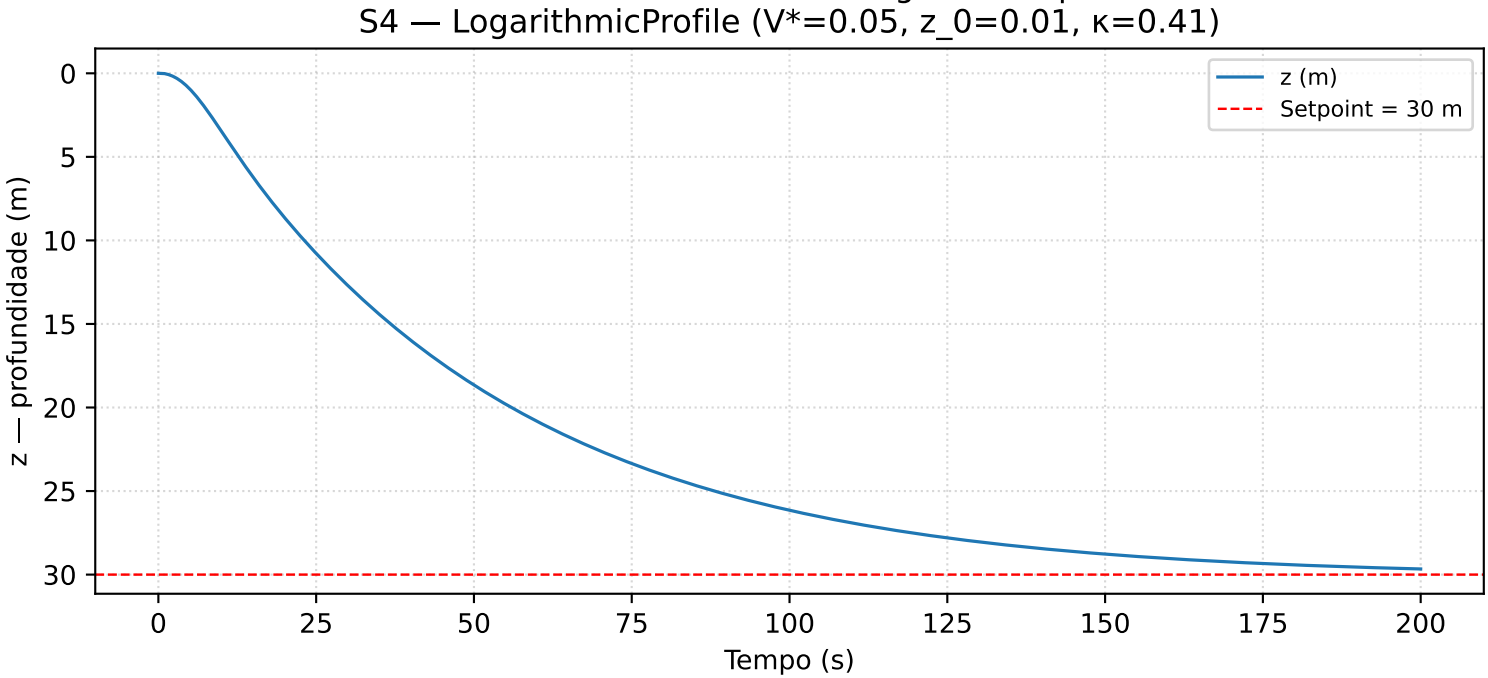

S3 — Profundidade ao longo do tempo
S3 — PowerLawProfile 1/7 (V_surface=1.0, z_ref=50)
z (m)
Setpoint = 30 m
z — profundidade (m)
Tempo (s)
0
5
10
15
20
25
30
0
25
50
75
100
125
150
175
200

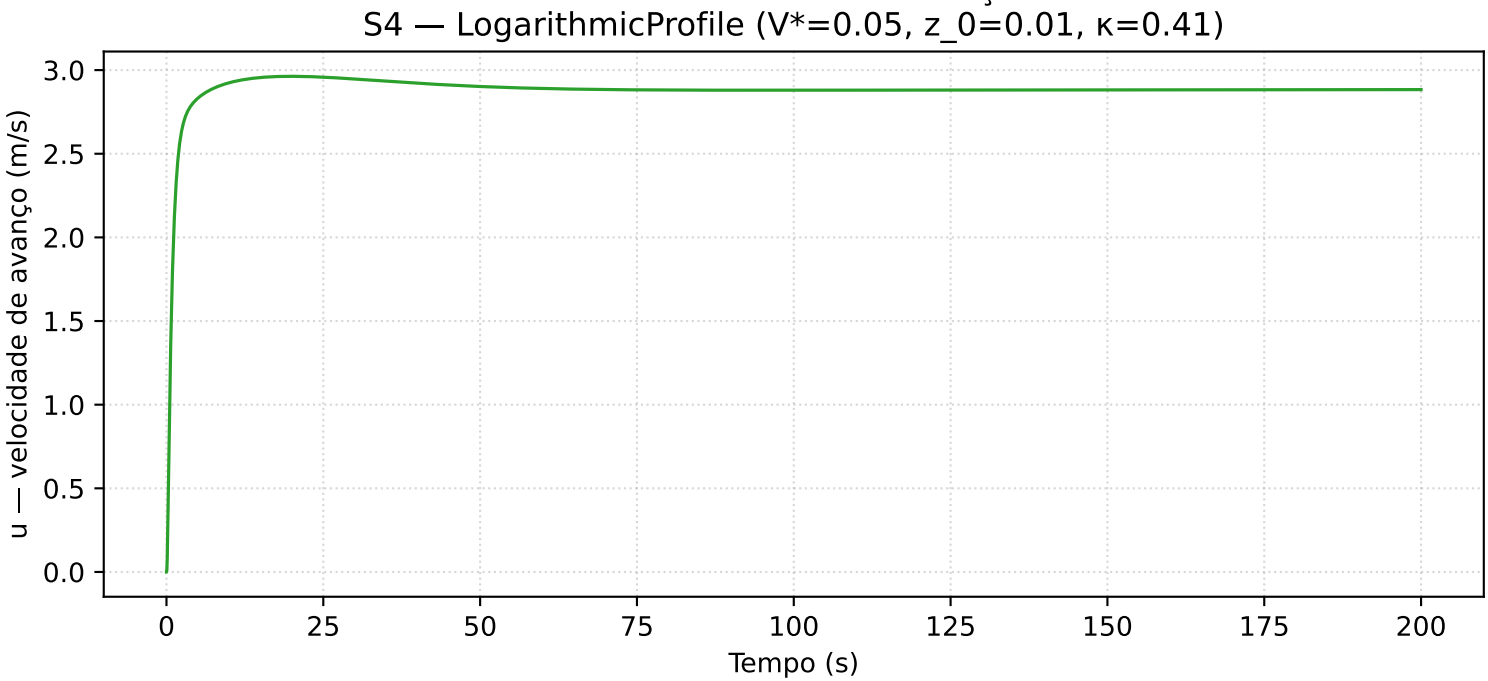

S3 — Velocidade de avanço
S3 — PowerLawProfile 1/7 (V_surface=1.0, z_ref=50)
u — velocidade de avanço (m/s)
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
25
50
75
100
125
150
175
200
Tempo (s)

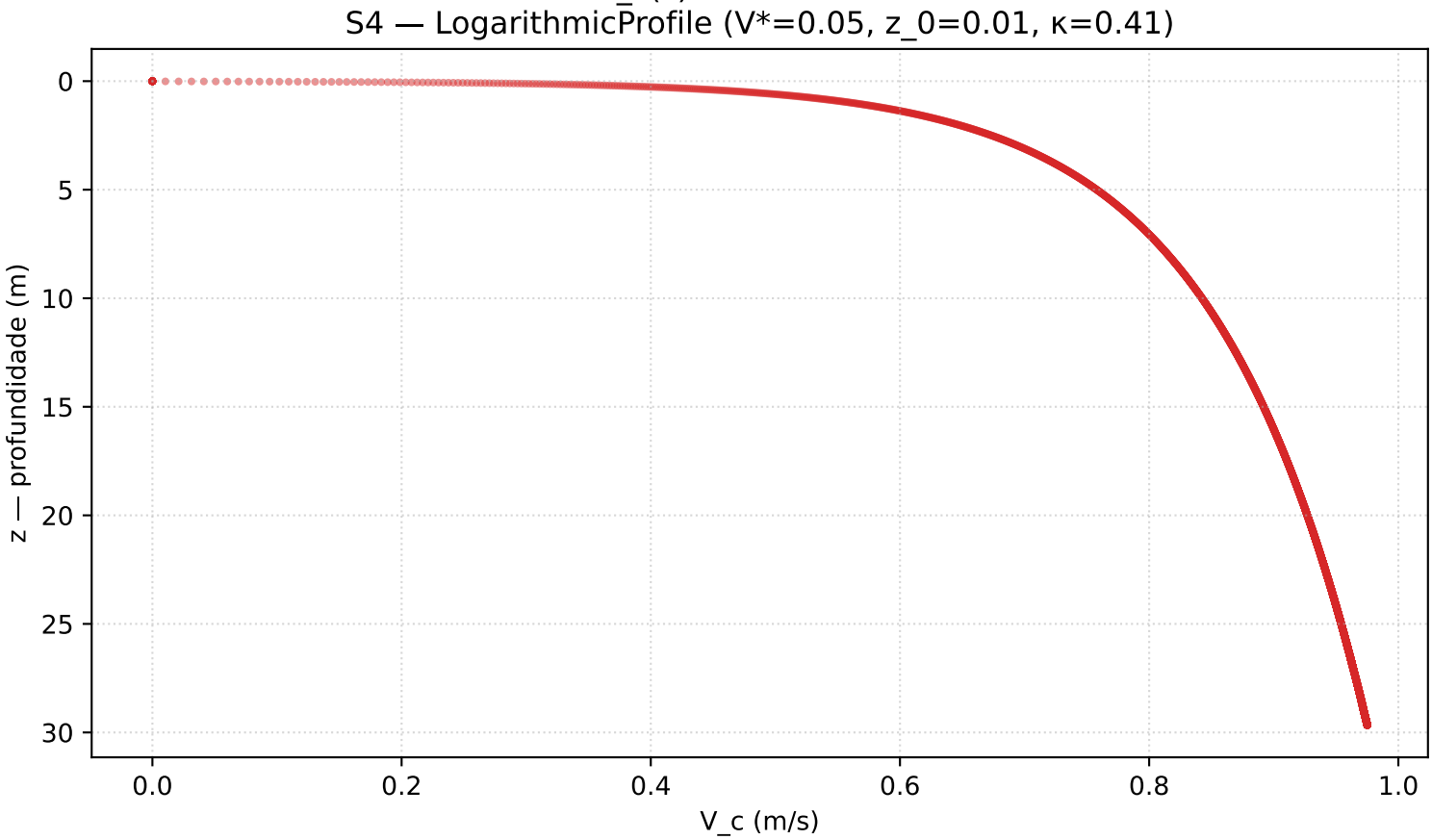

S3 — Perfil V_c(z) efectivo durante a missão
S3 — PowerLawProfile 1/7 (V_surface=1.0, z_ref=50)
z — profundidade (m)
0
5
10
15
20
25
30
0.0
0.2
0.4
0.6
0.8
V_c (m/s)

S4 — Trajectória 3D
S4 — LogarithmicProfile (V*=0.05, z_0=0.01, κ=0.41)

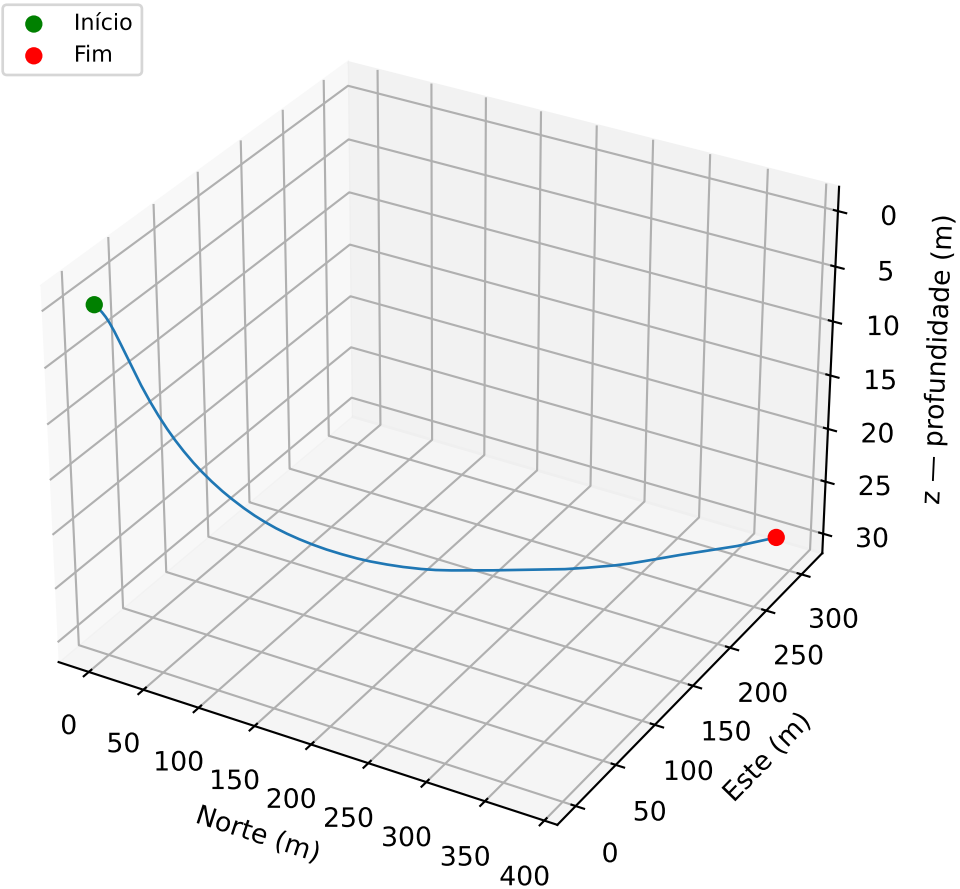


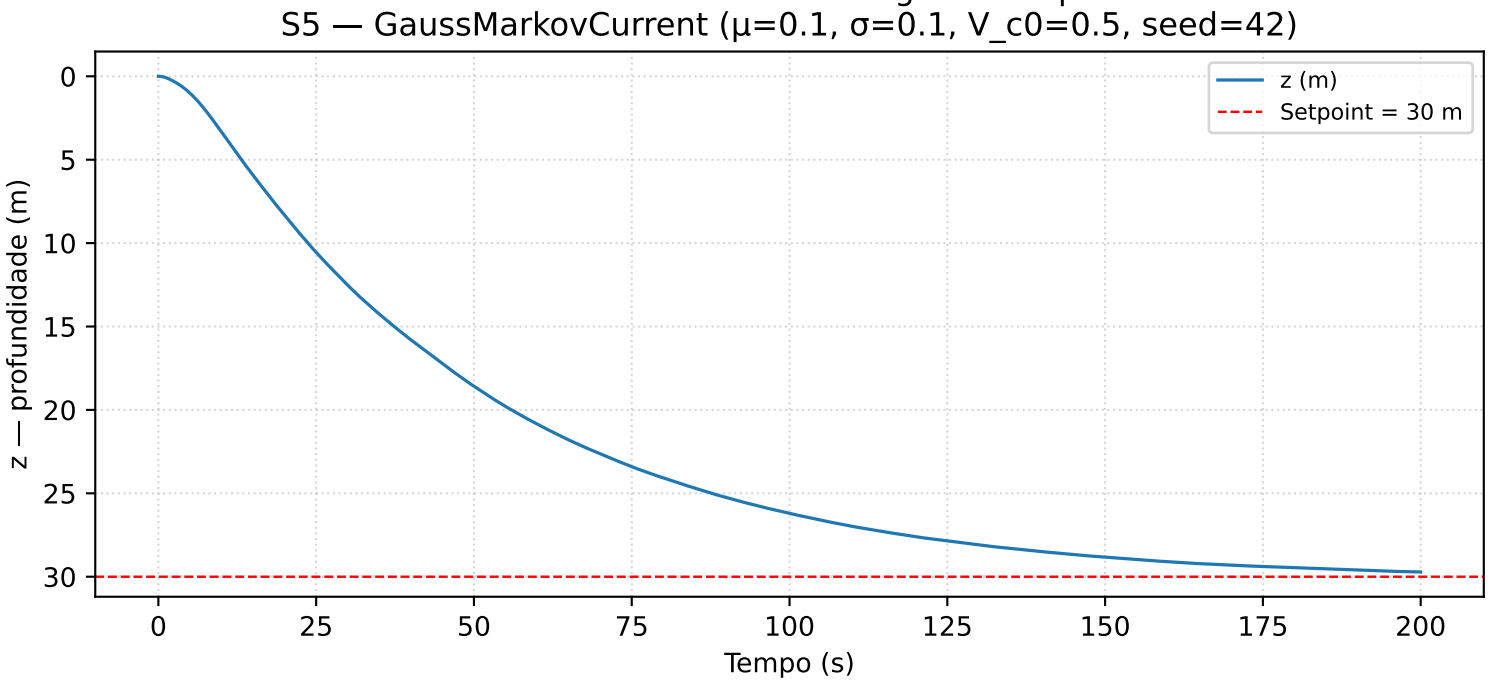

S4 — Profundidade ao longo do tempo
S4 — LogarithmicProfile (V*=0.05, z_0=0.01, κ=0.41)
z (m)
Setpoint = 30 m
z — profundidade (m)
Tempo (s)
0
5
10
15
20
25
30
0
25
50
75
100
125
150
175
200

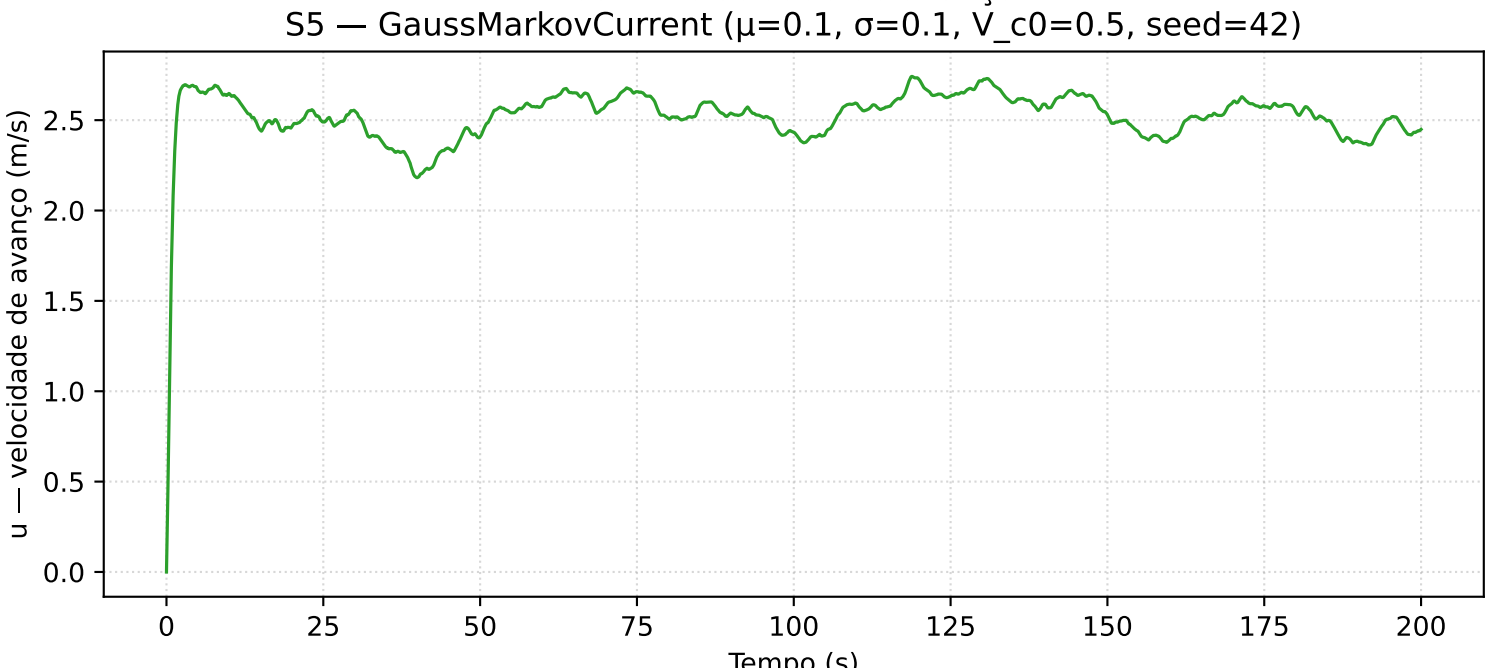

S4 — Velocidade de avanço
S4 — LogarithmicProfile (V*=0.05, z_0=0.01, κ=0.41)
u — velocidade de avanço (m/s)
Tempo (s)
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
25
50
75
100
125
150
175
200

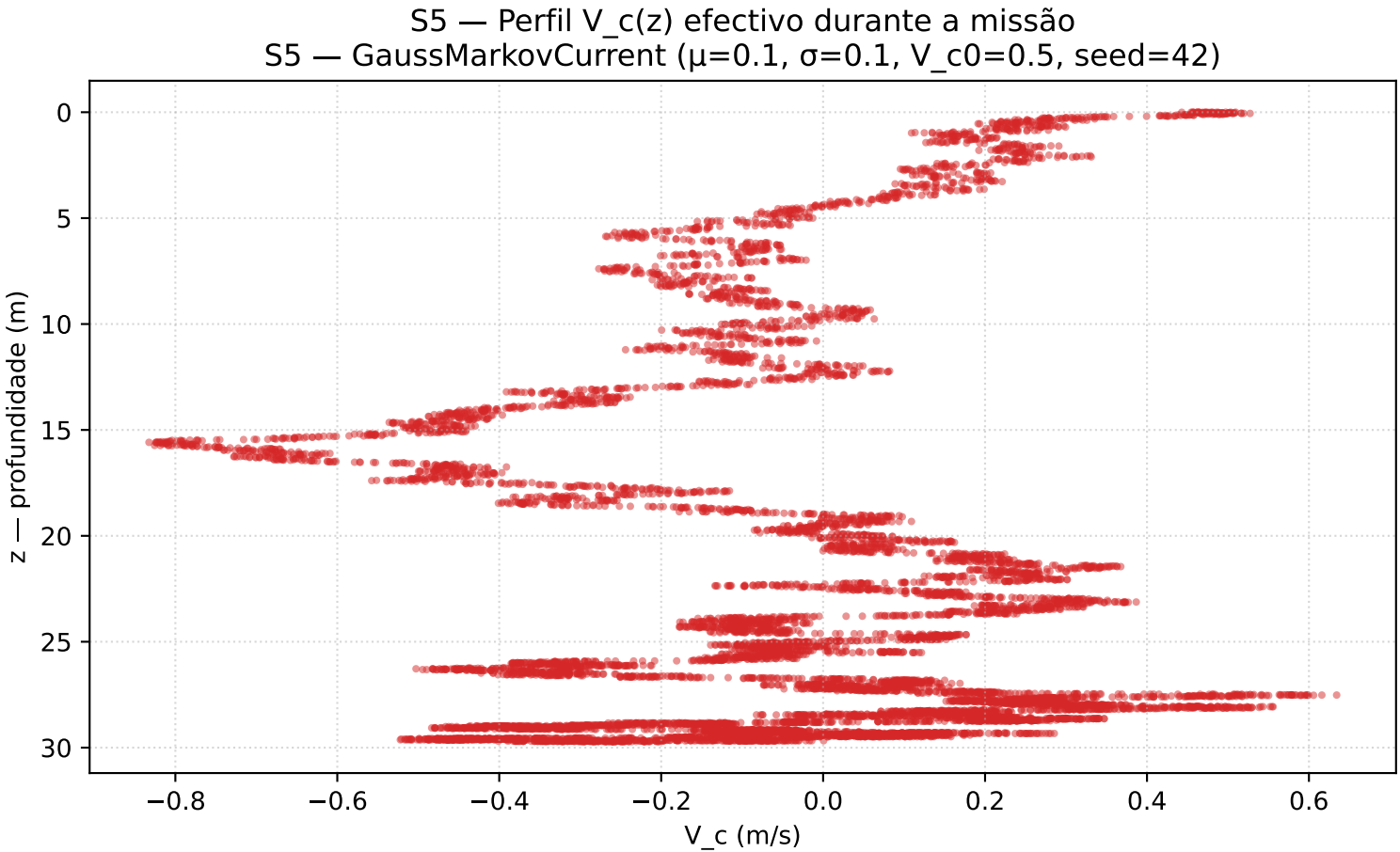

S4 — Perfil V_c(z) efectivo durante a missão
S4 — LogarithmicProfile (V*=0.05, z_0=0.01, κ=0.41)
z — profundidade (m)
V_c (m/s)
0
5
10
15
20
25
30
0.0
0.2
0.4
0.6
0.8
1.0

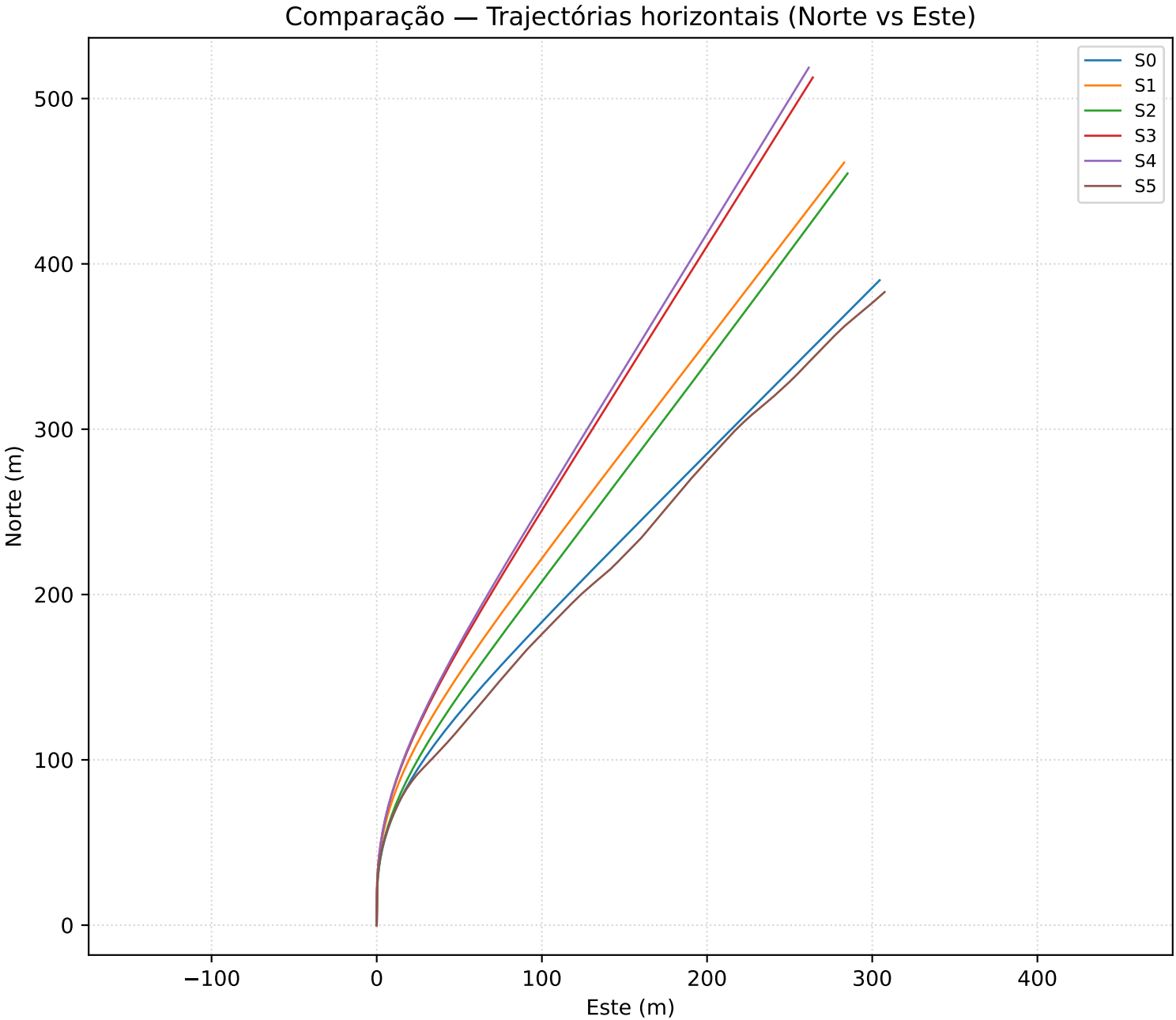
S5 — Trajectória 3D
S5 — GaussMarkovCurrent (μ=0.1, σ=0.1, V_c0=0.5, seed=42)

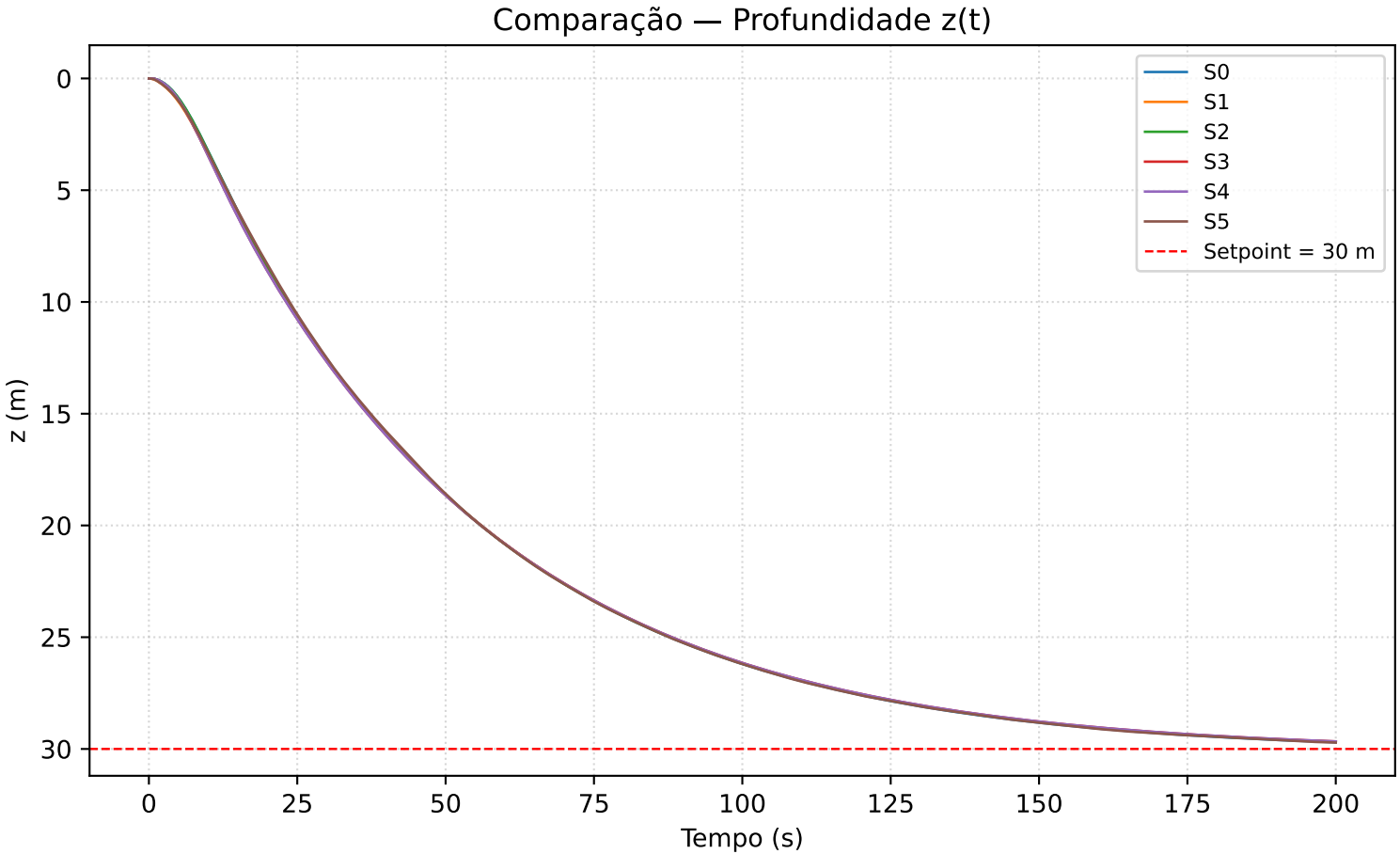


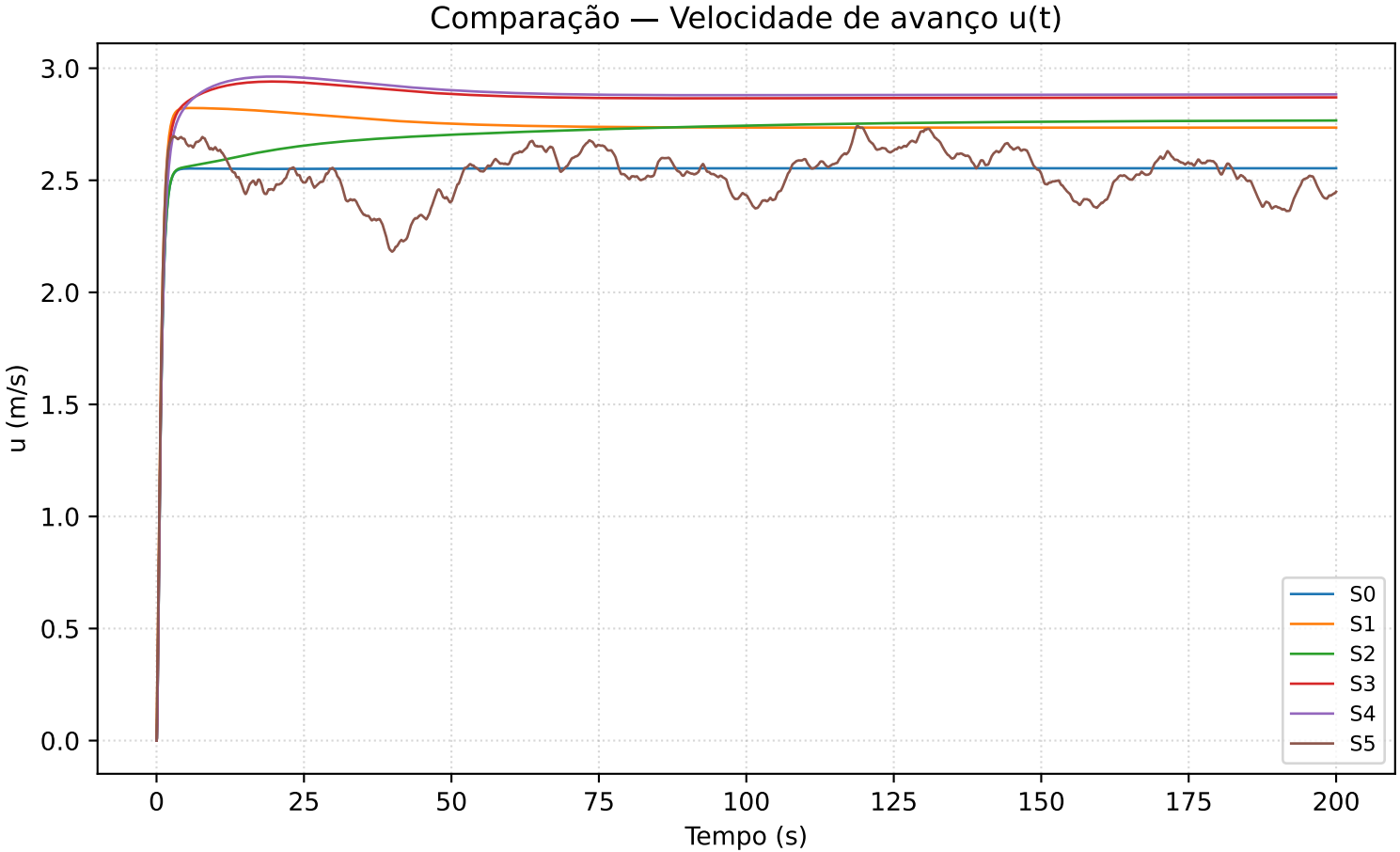

S5 — Profundidade ao longo do tempo
S5 — GaussMarkovCurrent (μ=0.1, σ=0.1, V_c0=0.5, seed=42)
z (m)
Setpoint = 30 m
z — profundidade (m)
Tempo (s)
0
5
10
15
20
25
30
0
25
50
75
100
125
150
175
200


S5 — Velocidade de avanço
S5 — GaussMarkovCurrent (μ=0.1, σ=0.1, V_c0=0.5, seed=42)
u — velocidade de avanço (m/s)
2.5
2.0
1.5
1.0
0.5
0.0
0
25
50
75
100
125
150
175
200
Tempo (s)


S5 — Perfil V_c(z) efectivo durante a missão
S5 — GaussMarkovCurrent (μ=0.1, σ=0.1, V_c0=0.5, seed=42)
z — profundidade (m)
V_c (m/s)
0
5
10
15
20
25
30
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6


Comparação — Trajectórias horizontais (Norte vs Este)
S0
S1
S2
S3
S4
S5
Norte (m)
Este (m)
0
100
200
300
400
500
−100
0
100
200
300
400


Comparação — Profundidade z(t)
S0
S1
S2
S3
S4
S5
Setpoint = 30 m
z (m)
0
5
10
15
20
25
30
0
25
50
75
100
125
150
175
200
Tempo (s)


Comparação — Velocidade de avanço u(t)
u (m/s)
Tempo (s)
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
25
50
75
100
125
150
175
200
S0
S1
S2
S3
S4
S5